

Signaling Safety

Teaching note

Michaely, Rossi, and Weber (2021), *Journal of Financial Economics*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

■ Dividend signaling was stuck for forty years

Miller-Modigliani (1961) plant the information-content idea; Bhattacharya (1979), Miller-Rock (1985), and John-Williams (1985) formalize it with a first-moment prediction: dividend changes should be followed by same-direction changes in cash flows. The empirical evidence consistently failed to support the prediction: Watts (1973) finds the relation trivially small, and Benartzi-Michaely-Thaler (1997) find dividend changes tied to past, not future, earnings.

“We conclude that managerial signaling motives [...] have at best minor influence on payout policy.”

DeAngelo et al. (2009, p. 95), as quoted in Michaely et al. (2021, p. 405)

The residual puzzle: announcement returns are large and managers affirm that payout conveys information (Brav et al. 2005), yet the predicted earnings changes never appear.

The tests measured realizations, not expectations

The theory concerns managers' private information about the distribution of future cash flows; the tests regressed realized accounting earnings changes on dividend changes. Realized earnings arrive at low frequency, are nonstationary in levels, and are manipulable in the short run (pp. 406, 408), with no consensus detrending adjustment (p. 408); total return volatility cannot separate cash-flow from discount-rate variation. The second-moment reading was latent in the evidence: Grullon-Michaely-Swaminathan (2002) document falling betas after dividend increases, and Lintner's managers raise dividends only on earnings changes they view as permanent.

"In this paper, we show dividends do convey information, but it is information about the second moment of earnings and not about the first moment."

Michaely, Rossi, and Weber (2021, p. 406)

The unsticking move: borrow the Module 7 design

Campbell (1991) built the aggregate news decomposition; Vuolteenaho (2002) carried it to the firm level; this paper carries it into event windows around payout decisions, where the object becomes the change in $\text{var}(\eta_{cf}) = (e1' + \lambda')\Sigma(e1 + \lambda)$ (eq. 11, p. 409) around the event.

“Because this method has so far not been applied in a corporate finance context, we briefly review the basic ingredients and closely follow the original notation.”

Michaely, Rossi, and Weber (2021, p. 408)

The stated advantages (p. 406): returns-based (high frequency, resistant to manipulation, stationary); separates cash-flow from discount-rate variation; yields novel evidence on discount rates and corporate decisions (Cochrane 2011).

Two VARs per event operationalize the design

For each event, one VAR is estimated on the 60 months before and one on the 60 months after: quarterly frequency; state vector of market-adjusted log return, log book-to-market, and log ROE; one lag; $\rho = 0.986$. The headline statistic is the change in $\text{var}(\eta_{cf})$, scaled by the mean pre-event variance.

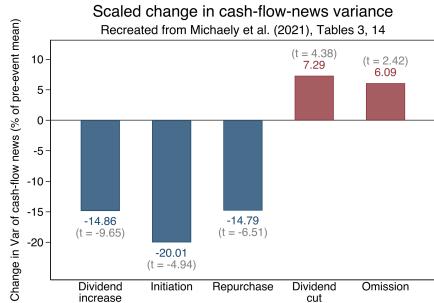
Event sample	N	Period
Dividend increases	2,441	1964-2013
Dividend decreases	2,461	1964-2013
Initiations	1,069	1964-2013
Omissions	1,233	1964-2013
Repurchase announcements	2,662	1980-2013

Screens follow Grullon-Michaely-Swaminathan (2002) and Michaely-Thaler-Womack (1995); omissions hand-verified against the WSJ Index; events at the same firm at least ten years apart; 10-34 post-event delistings within five years (p. 410).

Payout changes forecast cash-flow volatility

Cash-flow-news variance falls after the events that signal safety and rises after the events that signal risk (Tables 3 and 14).

The pattern survives alternative Γ restrictions, lag orders, window lengths, subperiods, and matched controls (Tables 4-9); placebo pre-event windows are “either statistically indistinguishable from zero or have the ‘wrong’ sign” (p. 415).



The nulls carry half the message

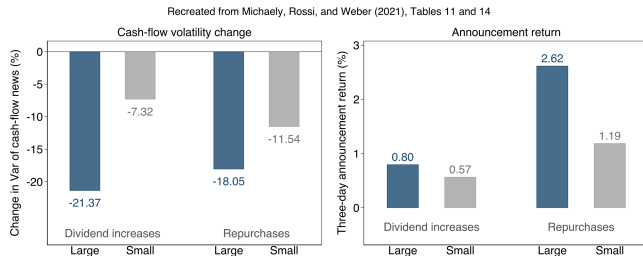
No first-moment change. The level of cash-flow news does not move around any event type (Table 3 Panel A; all t -statistics between -1.08 and 1.22): the old refutations are reproduced within the new design's own data.

No discount-rate change. Discount-rate news is approximately zero around every event type (Table 3 Panel B). Discount rates, the dominant force in aggregate asset pricing (Cochrane 2011), appear to play no role in payout decisions; cash-flow news drives the corporate policy.

No total-volatility change. Annualized stock return volatility does not change around dividend events (Online Appendix Table A.5), so total return volatility is a poor proxy for cash-flow volatility; only the decomposition isolates the moving part.

The three nulls are what the borrowed design adds: without the decomposition, none of these channels could be separated.

Larger signals are followed by larger responses



Top-tercile dividend increases are followed by a 21.37 percent volatility decline against 7.32 percent for bottom-tercile increases (difference $t = -12.30$, Table 11), with larger three-day announcement returns; large repurchase programs show the same ordering (Table 14). Larger signals carry more information.

Foregone investment makes the signal credible

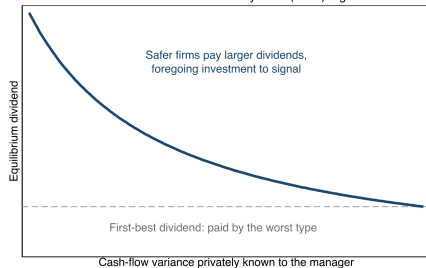
A Miller-Rock-style model (Section 5): the manager privately learns cash-flow variance; production is concave, so higher variance lowers expected output.

The separating equilibrium has dividends decreasing in variance: safer firms pay more and forego investment.

Predictions 1 and 3 restate the time-series and announcement-return facts; Prediction 2 discriminates.

Safer firms pay larger dividends in equilibrium

Illustration based on Michaely et al. (2021) Fig. 2



The cross-section sides with signaling

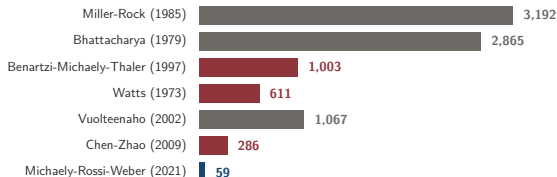
Because the cost of the signal is foregone investment, a dollar of dividends is more informative when current earnings are smaller, so signaling predicts a positive interaction between the dividend change and the earnings level; additive agency predicts a negative one (Section 5.2).

Regressor (Table 12, col. 3)	Coefficient	<i>t</i>
Δ Dividend	-0.37	-5.94
Earnings per share	-0.12	-1.87
Δ Dividend $\times$ eps	0.24	3.12

The interaction is positive: the same dollar of dividends is followed by a 25 percent larger volatility reduction for firms with smaller current earnings (p. 406), and the effect concentrates in financially constrained firms (Hadlock-Pierce index). Heterogeneity by idiosyncratic volatility and book-to-market points the same way (Tables 15-16).

The new design arrived without its critique

The headline outcome is the variance of a residual, and Chen-Zhao (2009), published twelve years earlier, showed that residual absorbs VAR misspecification; neither Chen-Zhao (2009) nor Chen-Da-Zhao (2013) is cited (confirmed against the published reference list). Discussion question: does differencing the same firm's news variance across the event remove stable misspecification, and what assumption would make that argument rigorous?



Citations (OpenAlex, Aug. 2026), journal records

Gray: theory and borrowed technology; maroon: refutations and the uncited critique; navy: the focal paper. Only about one in six indexed works citing Bhattacharya (1979) also cites Benartzi-Michaely-Thaler (1997).

A new design can get a literature unstuck

What unstuck the literature was neither new data nor a new theory but a new instrument: the question was four decades old, and the answer changed when the design changed, from regressing realized earnings on dividend changes to measuring the market's expectations, and their volatility, directly from returns.

The seminar's running parallel: Dechow (1994) imported the Vuong test, Vuolteenaho (2002) imported the Campbell decomposition, and this paper imports Vuolteenaho; each borrowed design made a previously untestable comparison testable.

Openings for accounting research. The accounting adopters (Ham-Kaplan-Utke 2023; Kapons et al. 2023, both RAST), on their indexed reference lists, cite the finding without the machinery; the first-moment rehabilitation of Ham-Kaplan-Leary (2020) remains unadjudicated against the second-moment reframing. Candidate designs: the second-moment test with realized accounting earnings volatility; payout signals in credit markets; accrual-quality interactions with the volatility signal.

Summary

- Michaely, Rossi, and Weber confront a literature stuck for four decades, in which theory predicted earnings changes that never appeared, and get it unstuck by importing a design that measures what the theory is about: expected cash flows, and their volatility, extracted from returns.
- The evidence the new design produces, volatility falling after increases and repurchases and rising after cuts and omissions, with nulls on the first moment and on discount-rate news, is consistent with payout conveying information about safety; the cross-sectional test favors signaling over agency.
- The transplant carries the seminar's recurring caution: the imported measure is the variance of a residual, the critique of that construction did not travel with it, and a design that unsticks a literature warrants the same methodological scrutiny as the designs it replaces.

Michaely-Rossi-Weber (2021) · Vuolteenaho (2002) · Benartzi-Michaely-Thaler (1997) ·
Chen-Zhao (2009) · github.com/sbb151/acctg597a-research-design